

# Computer Networks and Data Communication

## Course Code: CSC214

## LEC 1 : Introduction to Communication Networks



## How Your 100 Marks Are Distributed



**Plan your effort across the semester — every component matters.**

## Coursework Breakdown

10

**Sections**

marks

10

**Lectures**

marks

10

**Quizzes**

marks

10

**Reports**

marks

**Stay consistent: coursework is 40% of your final grade.**

## Final Exam Structure

### Electronic Part

**16**

marks

Approximately 40% of the final exam

### Written Paper

**24**

marks

60% of the final exam

**Total final exam = 16 + 24 = 40 marks**

KEEP THESE TWO CONDITIONS IN MIND

## Passing Requirements

Overall Course Grade

**60 / 100**

Minimum passing grade

Final Exam Minimum

**12 / 40**

Required in the final exam

**Both requirements must be satisfied to pass the course.**

# CSC214 | COURSE CONTENTS

From network fundamentals to data transmission technologies

## 01 NETWORK FUNDAMENTALS

- Networking & data transmission.
- Classifications Topologies.
- Architecture Applications

## 02 NETWORK MODELS

- OSI & TCP/IP
- Protocols & Standards
- Communication models

## 04 DATA TRANSMISSION

- Transmission techniques
- Transmission media
- Analog & digital transmission

## 05 TRANSMISSION TECHNOLOGIES

- Optical fiber systems
- Modems & modulation
- Impairments Encoding Multiplexing

# DATA COMMUNICATIONS

*The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.*

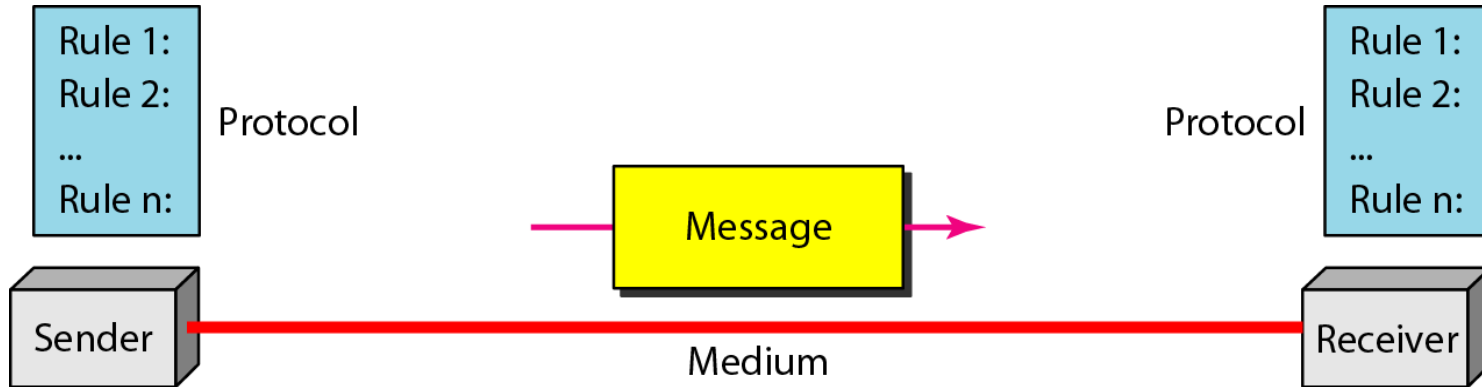
# Components of a data communications system

- **Message**
  - Information (data) to be communicated.
  - Such as text, numbers, pictures, audio, and video.
- **Sender**
  - Device that sends the data message.
  - Computer, workstation, and video camera
- **Receiver**
  - Device that receives the message.
  - Computer, workstation, and telephone handset.
- **Transmission medium**
  - Physical path by which a message travels from sender to receiver.
  - Twisted pair wire, coaxial cable, and satellite.
- **Protocol**
  - Set of rules that govern data communications.



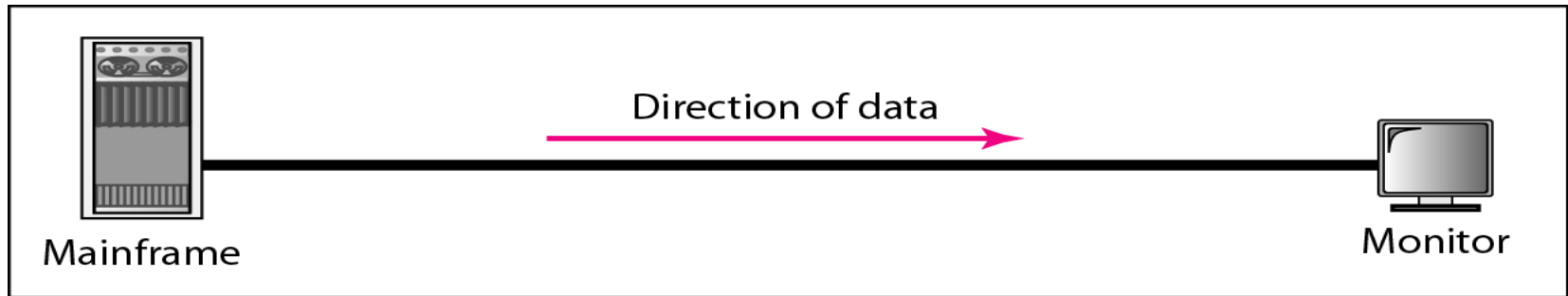
**Figure:** *Five components of data communication*

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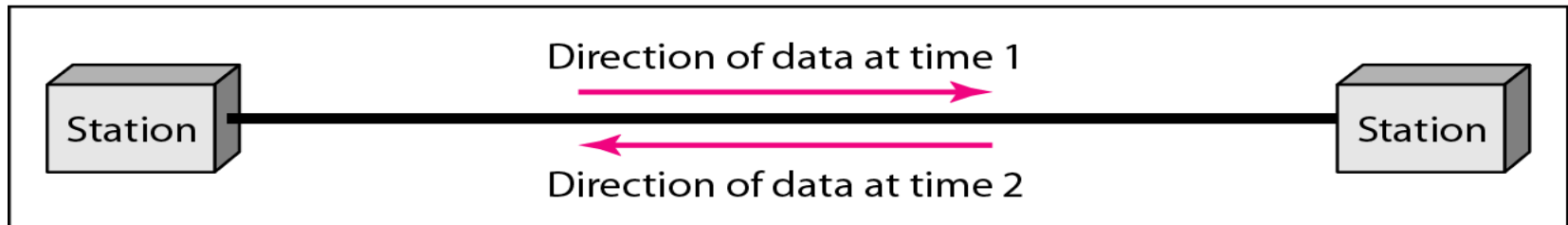


**Figure :** *Data flow (simplex, half-duplex, and full-duplex)*

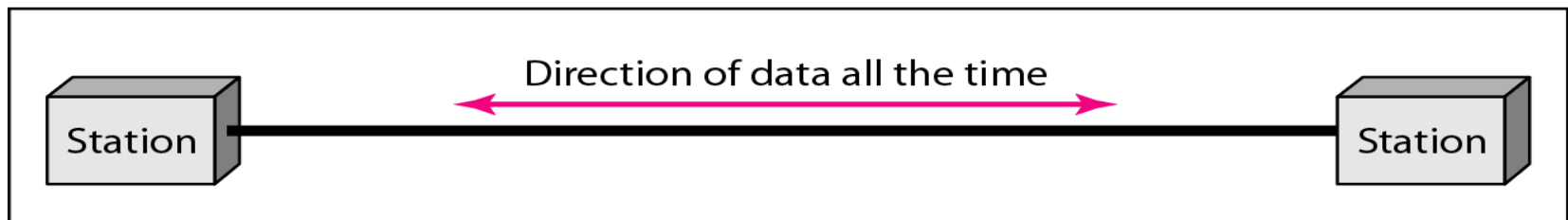
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a. Simplex



b. Half-duplex



c. Full-duplex

- Simplex

- The communication is unidirectional.
- Keyboard and monitors.

- Half-duplex

- Each station can both submit and receive, but not at the same time.
- Walkie-talkies

- Full-duplex

- Both stations can transmit and receive simultaneously.
- Telephone network.

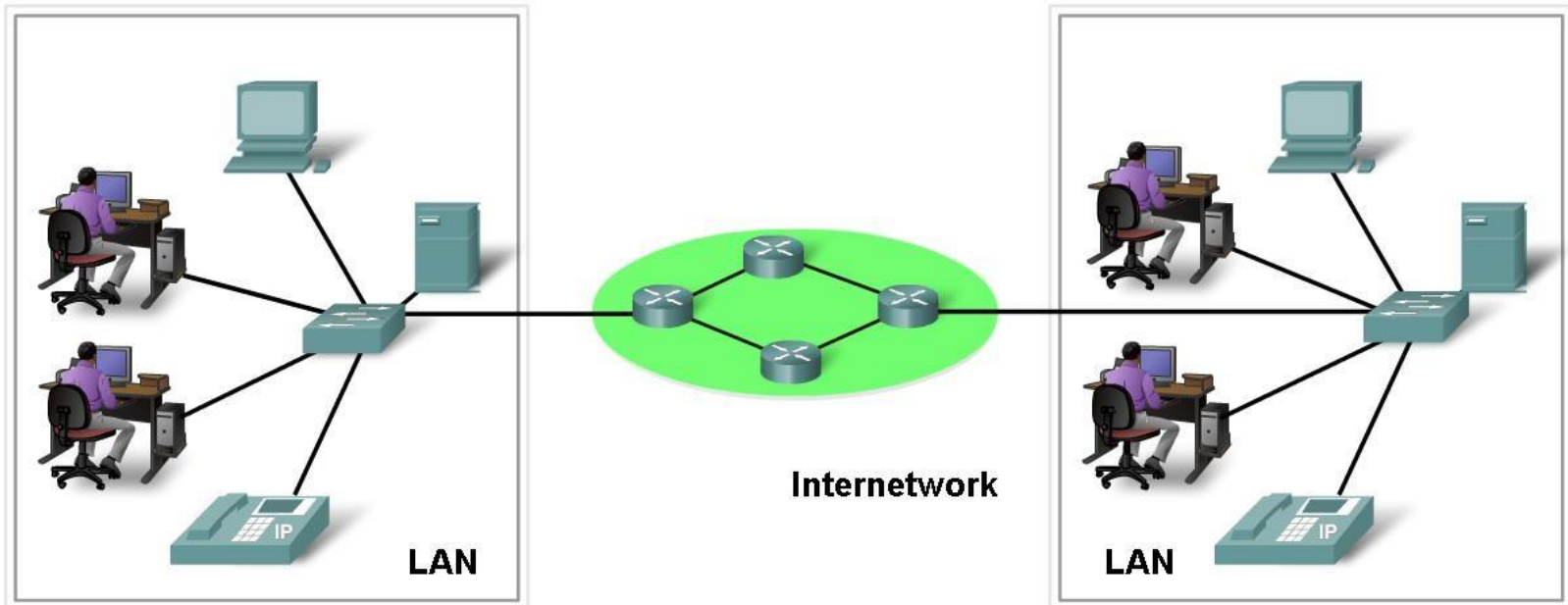
# NETWORKS



*A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.*

# What is the Computer Network?

- Computer Network means, a collection of **Computers, and other devices, or peripherals connected together through connecting media.**

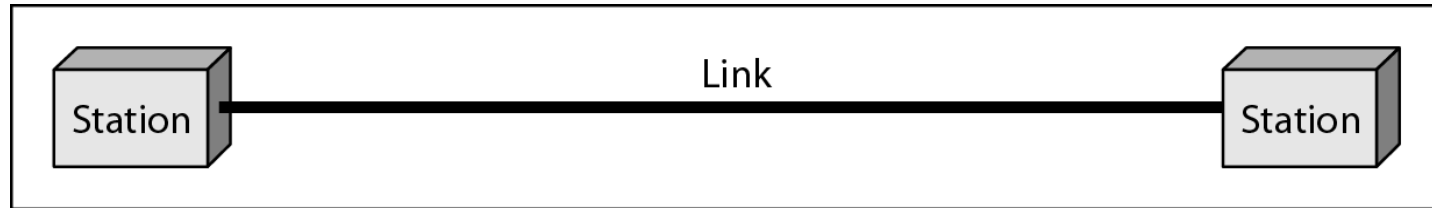


# Type of connection

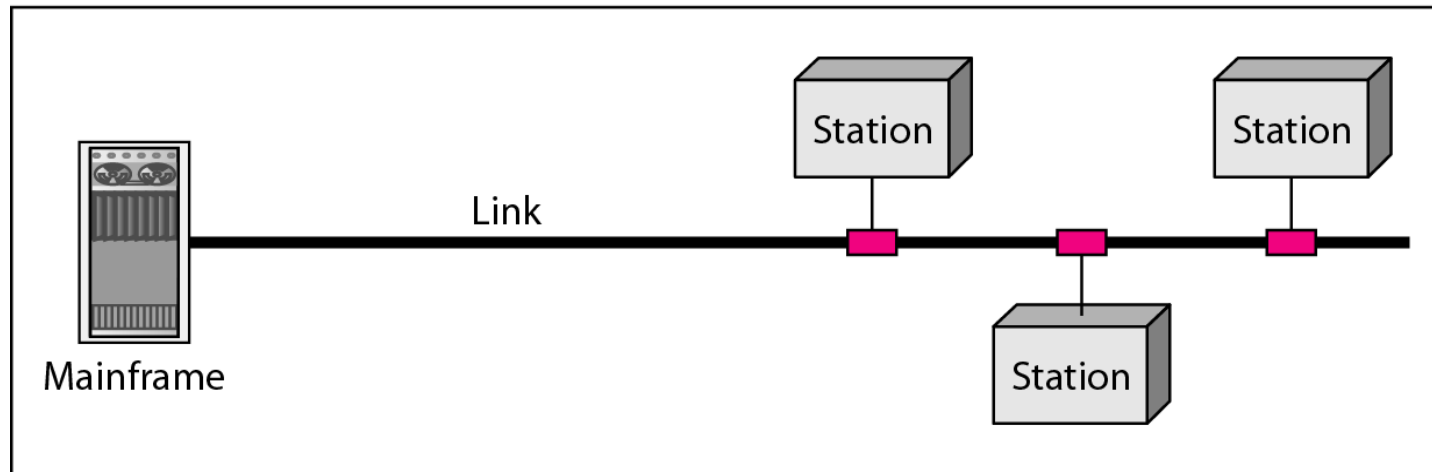
- Point-to-point
  - Dedicated link between two devices.
  - The entire capacity is reserved from transmission between those two devices.
- Multipoint (Multidrop)
  - More than two devices share a single link.
  - Can be spatially or time shared connection

**Figure 1.3** *Types of connections: point-to-point and multipoint*

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a. Point-to-point



b. Multipoint

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# **HUB vs SWITCH vs ROUTER**

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# HUB



## Layer 1 • Physical Layer

- Works at Layer 1 (Physical Layer)
- Broadcasts data to all ports
- No filtering or intelligent forwarding
- Half-Duplex only
- One collision domain for all ports
- Low cost and simple device
- Obsolete in modern networks



# SWITCH



## Layer 2 • Data Link Layer

- Works at Layer 2 (Data Link Layer)
- Forwards data to the correct port only
- Uses MAC address table
- Full-Duplex support
- Each port is a separate collision domain
- Higher performance and more secure
- Most common device in modern networks

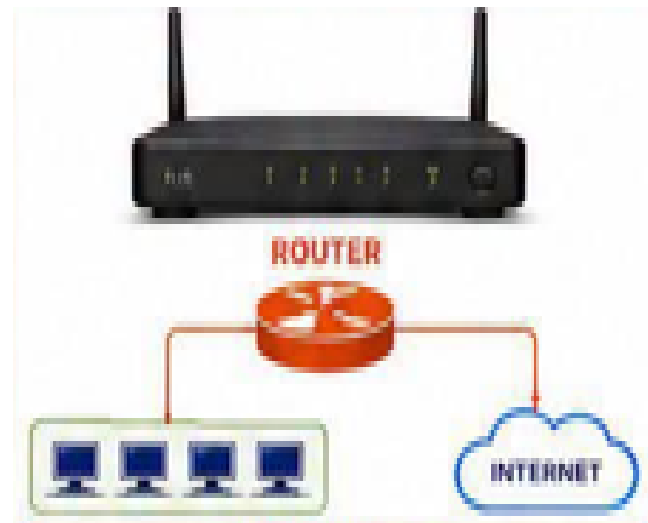


# ROUTER



## Layer 3 • Network Layer

- Works at Layer 3 (Network Layer)
- Routes data between different networks
- Uses IP address to find the best path
- Connects multiple networks (LAN to WAN)
- Breaks broadcast domains
- Essential for internet connectivity
- More complex and intelligent device



# HUB vs SWITCH vs ROUTER — AT A GLANCE

| Feature          | HUB             | SWITCH        | ROUTER            |
|------------------|-----------------|---------------|-------------------|
| OSI Layer        | Layer 1         | Layer 2       | Layer 3           |
| Primary Function | Repeats Signal  | Forwards Data | Routes Data       |
| Address Used     | None            | MAC Address   | IP Address        |
| Broadcast Domain | One (All Ports) | One per VLAN  | Separates Domains |
| Collision Domain | One (All Ports) | One per Port  | One per Interface |
| Duplex Mode      | Half-Duplex     | Full-Duplex   | Full-Duplex       |
| Performance      | Low             | High          | High              |
| Cost             | Very Low        | Moderate      | High              |
| Use Case         | Small, outdated | LAN, Offices  | Internet, WAN     |

## WHEN TO USE EACH DEVICE?

### HUB



Use a HUB only in small, simple networks where cost is the only concern. Almost never recommended today.

### SWITCH



Use a SWITCH in most LAN environments for better performance, security, and full-duplex communication.

### ROUTER



Use a ROUTER to connect different networks and access the internet. Essential for any multi-network setup.

# NETWORK TOPOLOGY

## Device Connection Structure

Network Topology is the physical layout of network components — how devices (computers, switches, routers, printers) are connected to each other and how data flows between them.

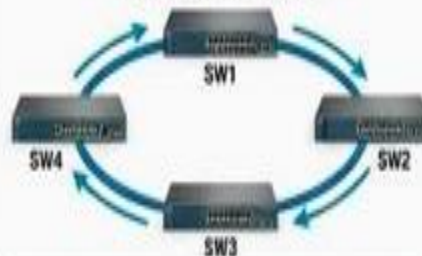
### STAR TOPOLOGY

CENTRALIZED CONNECTION



### RING TOPOLOGY

CIRCULAR CONNECTION



### MESH TOPOLOGY

MULTIPLE PATHS



HIGH REDUNDANCY

### BUS TOPOLOGY

SHARED BACKBONE



SINGLE BACKBONE

### HYBRID TOPOLOGY

COMBINED DESIGN



# STAR TOPOLOGY



## Centralized Connection

All devices connect to a central Managed Switch. The most common LAN design used in modern networks.

### Central Switch

Managed Switch acts as the hub of all connections

### Failure Isolation

One device failure doesn't bring down the network

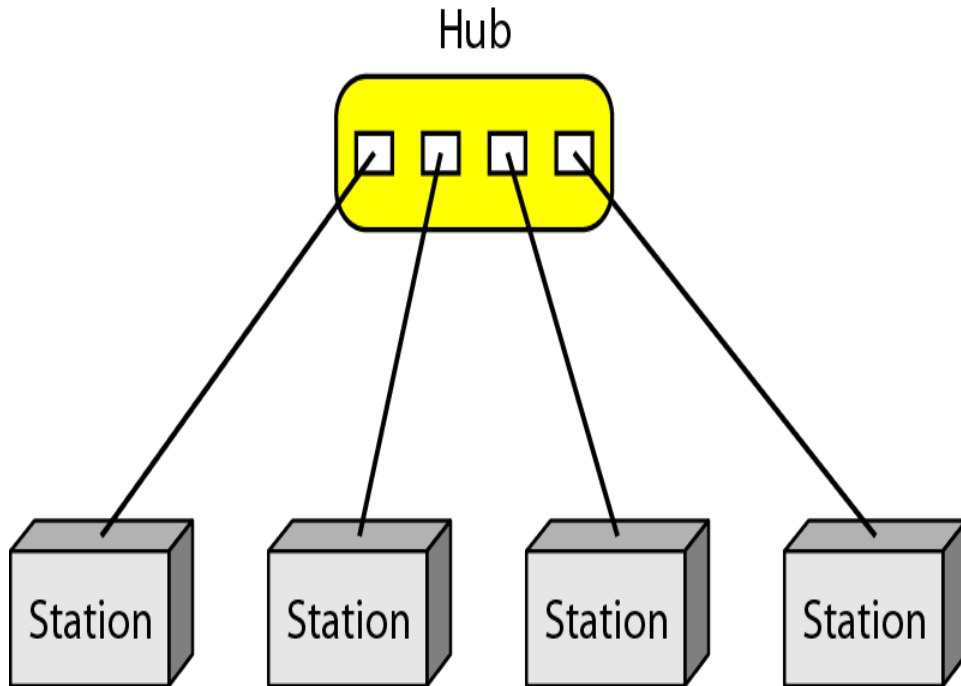
### Easy Management

Simple to add/remove devices without affecting others

### Most Common

Standard design for offices, schools & homes

**Figure:** *A star topology connecting four stations*





# BUS TOPOLOGY



## Shared Backbone

All devices share a single backbone cable. Data travels in both directions along the main line.

### Single Backbone

One main cable connects all devices in a linear fashion

### Simple & Cheap

Requires less cabling than star topology

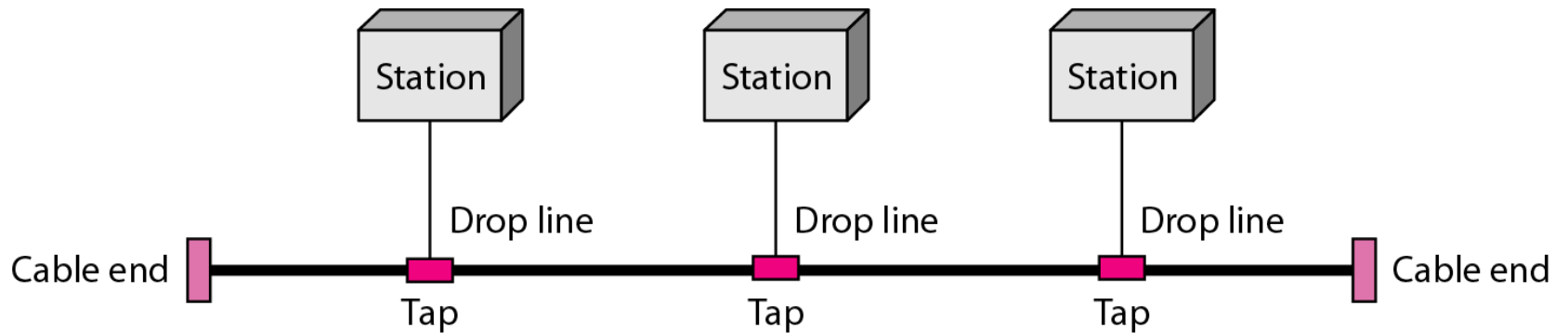
### Collision Domain

All devices share one collision domain — performance drops with more devices

### Single Point of Failure

If the backbone cable fails, the entire network goes down

**Figure:** A bus topology connecting three stations



# RING TOPOLOGY



## Circular Path Connection

Each device connects to exactly two others, forming a closed loop. Data travels in one direction around the ring.

### Circular Path

Data flows in a continuous loop from device to device

### Token Passing

Often uses token to control who can transmit data

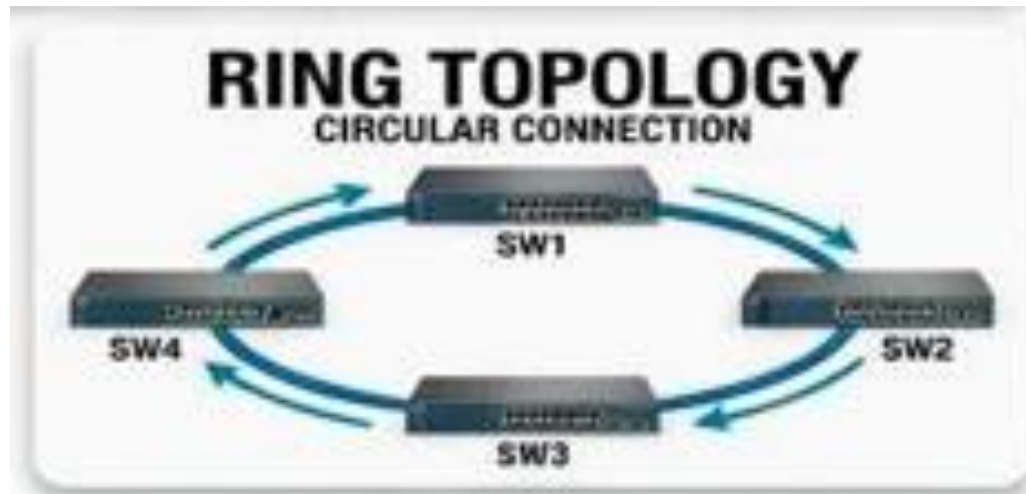
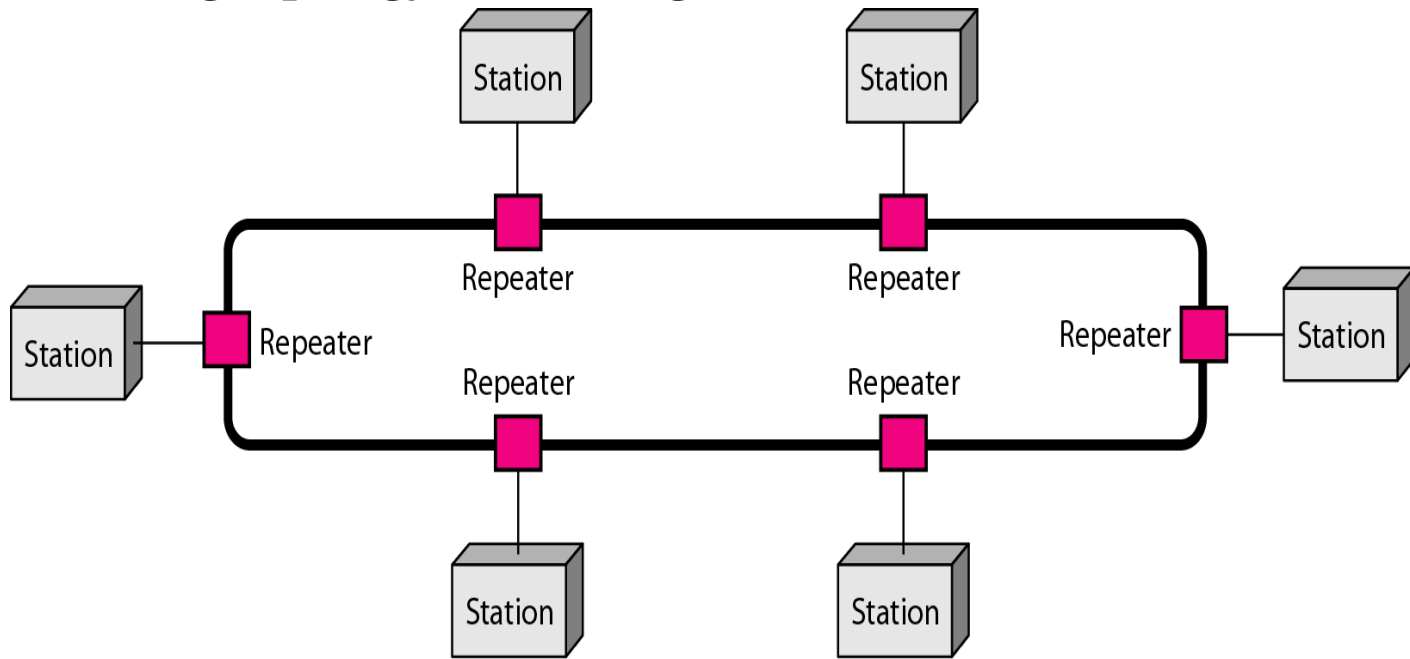
### Equal Access

Every device has equal opportunity to send data

### Single Failure Risk

One broken connection can disrupt the entire ring

**Figure:** *A ring topology connecting six stations*



# MESH TOPOLOGY



## Multiple Paths • High Redundancy

Every device connects to every other device (full mesh) or to multiple others (partial mesh). Maximum fault tolerance.

### High Redundancy

Multiple paths ensure data always finds a way

### Fault Tolerance

Network survives even if several links fail

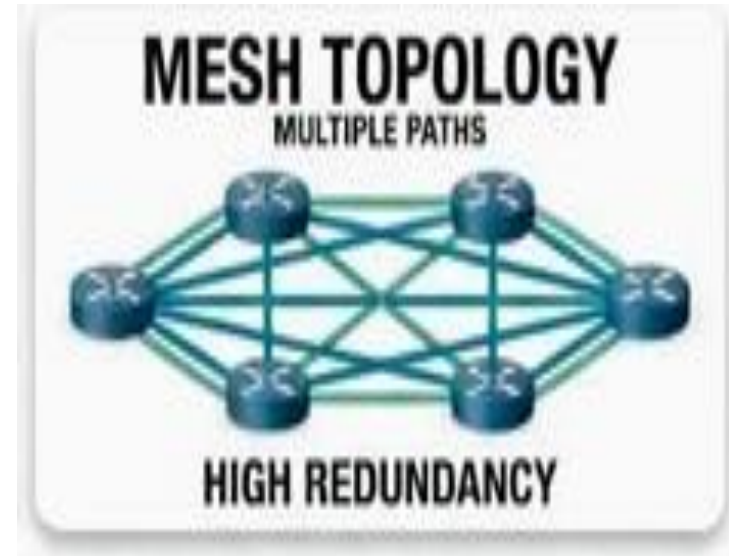
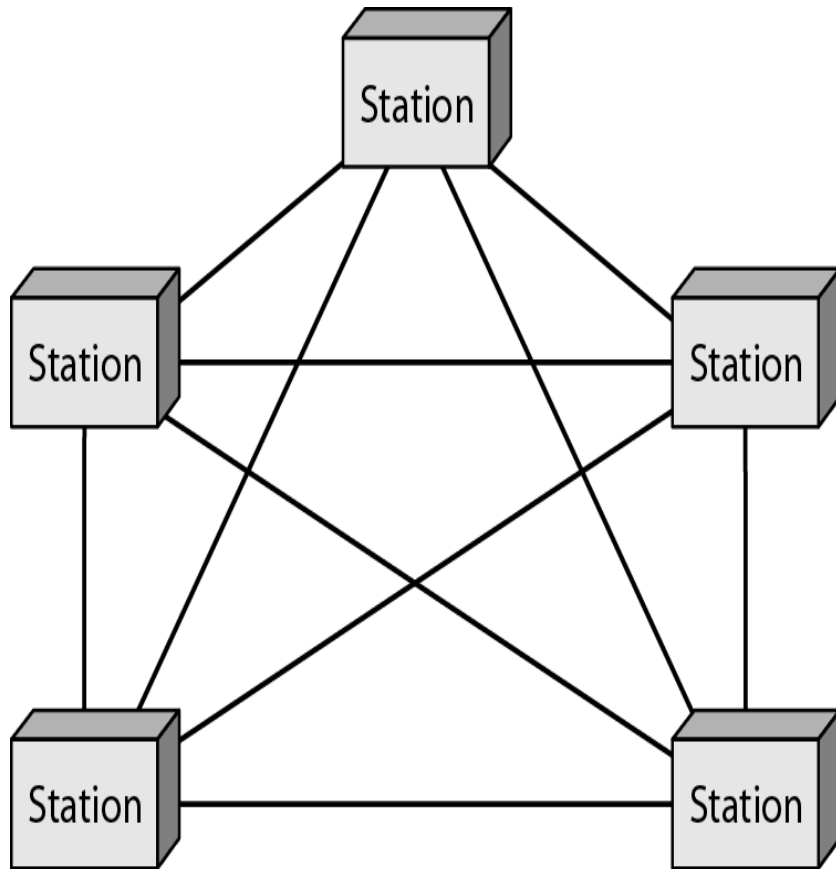
### Expensive

Requires many cables and complex configuration

### Best For Critical Systems

Used in data centers and mission-critical networks

**Figure:** *A fully connected mesh topology (five devices)*



# HYBRID TOPOLOGY



## Combined Design • Complex Enterprises

Combines two or more topologies (Star + Mesh, Star + Ring, etc.) to meet specific organizational needs.

### Flexible Design

Mix topologies to fit different departments or sites

### Scalable

Grow the network by adding new topology segments

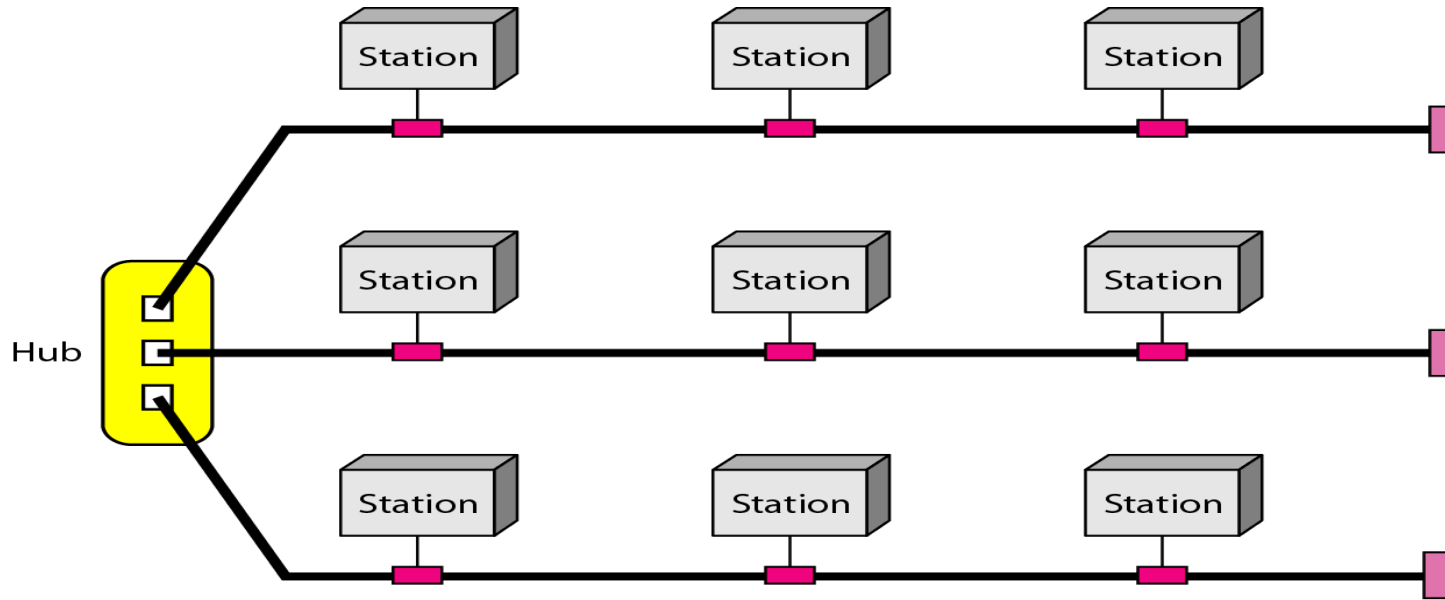
### Real Enterprise Use

Core Network + Branch Networks + Access Layers

### Complex Management

Requires skilled admins and careful planning

**Figure:** *A hybrid topology: a star backbone with three bus networks*





# TOPOLOGY COMPARISON

| Topology      | Structure              | Advantages                                                                               | disadvantages                                                                                                   |
|---------------|------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>STAR</b>   | <b>Central Switch</b>  | Easy to manage<br>* Easy to expand<br>* Device failure isolated<br>* Centralized control | 1. High cost<br>2. Central switch fails = all down<br>3. Needs more cables                                      |
| <b>BUS</b>    | <b>Shared Backbone</b> | Simple to install<br>* Cheap<br>* Less cabling<br>* Easy to extend                       | 1. Cable break = network down<br>2. One collision domain<br>3. Slow with many devices<br>4. Hard to find faults |
| <b>RING</b>   | <b>Circular Path</b>   | Equal access for all<br>* No collisions<br>* Predictable speed<br>* Simple data path     | 1. One break stops all<br>2. Hard to add devices<br>3. Slow with large size.                                    |
| <b>MESH</b>   | <b>Multiple Paths</b>  | Highest redundancy<br>* Multiple paths<br>* Very fault tolerant<br>* Secure & reliable   | 1. Very expensive.<br>2. Complex setup.<br>3. Needs many cables.                                                |
| <b>HYBRID</b> | <b>Combined Design</b> | Flexible design<br>* Scalable<br>* Best of all types<br>* Fits large companies           | 1. Hard to design.<br>2. Expensive to manage.<br>3. Needs skilled admins.                                       |

# Categories of networks

## LAN vs. MAN vs. WAN

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| Category   | Typical coverage                  | Example environment     | Key idea                   |
|------------|-----------------------------------|-------------------------|----------------------------|
| <b>LAN</b> | Single office / building / campus | University building     | Local resource sharing     |
| <b>MAN</b> | Town / city                       | City-wide network       | Connects multiple LANs     |
| <b>WAN</b> | Country / continent / world       | ISP / intercity network | Long-distance connectivity |

Local Area Network: (LAN)

Metropolitan Area Network: (MAN)

Wide Area Network: (WAN)



**firewall**  
جدار الحماية



**Firewall**



**internet**

# Firewall

Software hardware

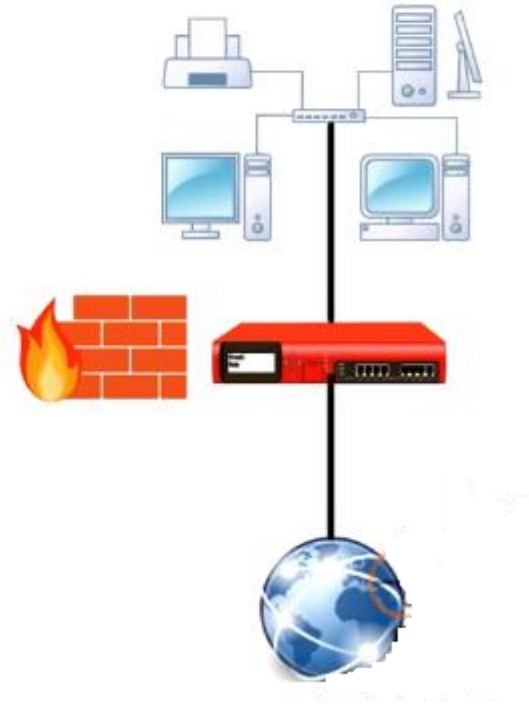
## Host-based Firewall



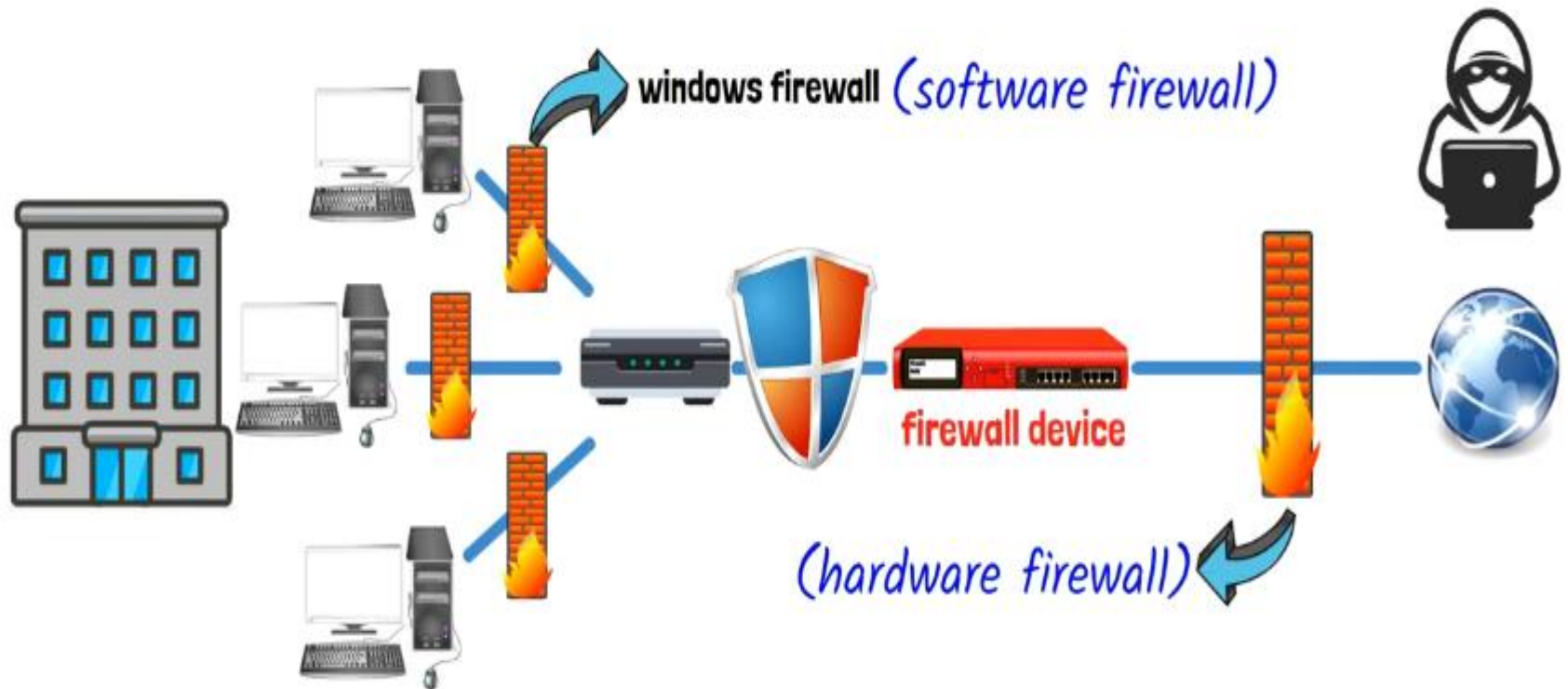
by default



Windows



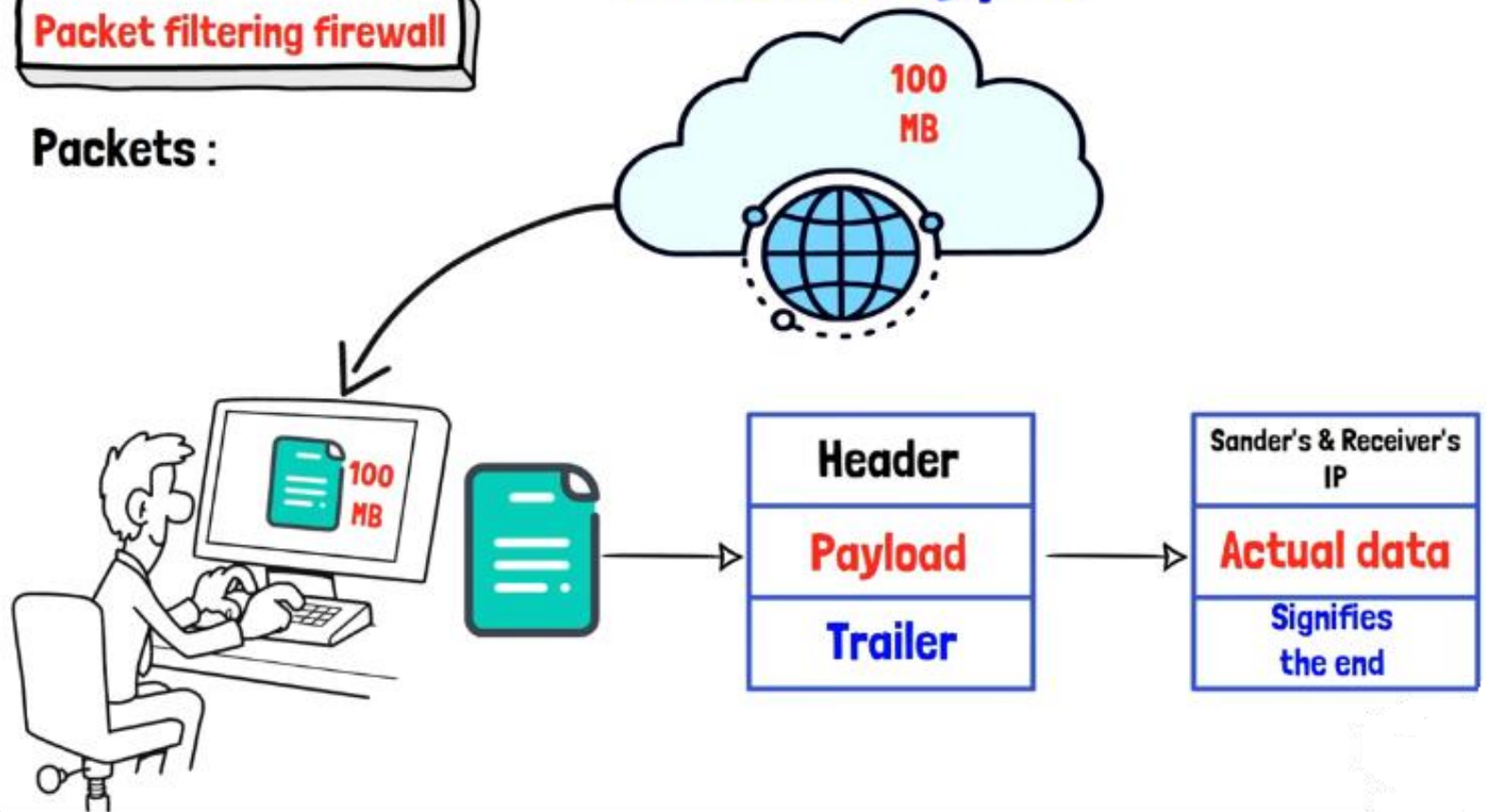
## Network-based firewall :



# Firewall types

## Packet filtering firewall

Packets :



# Firewall types

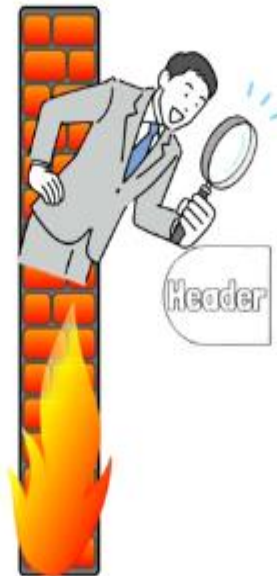
Packet filtering firewall



|             |
|-------------|
| IP Sender   |
| IP Receiver |
| Port num    |



Access control  
list



Header Payload Trailer

Packet filtering firewall

Port number :

**192.168.1.1:8080**



# Firewall types

Packet filtering firewall

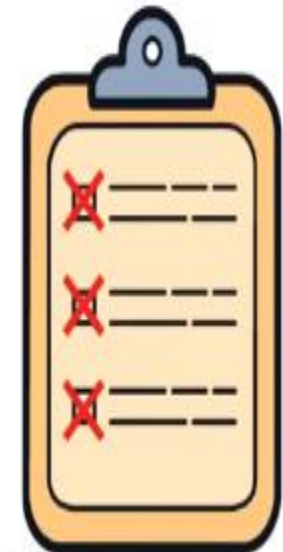


Simple

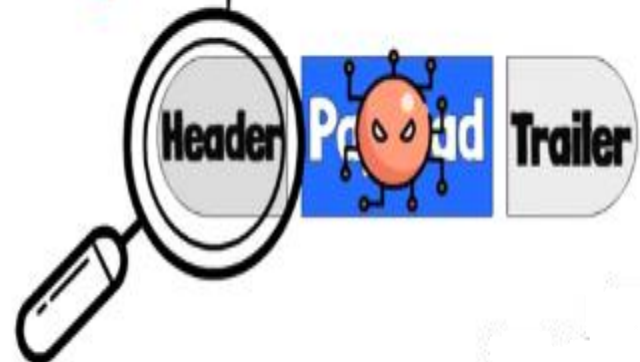


Low security

|             |
|-------------|
| IP Sender   |
| IP Receiver |
| Port num    |



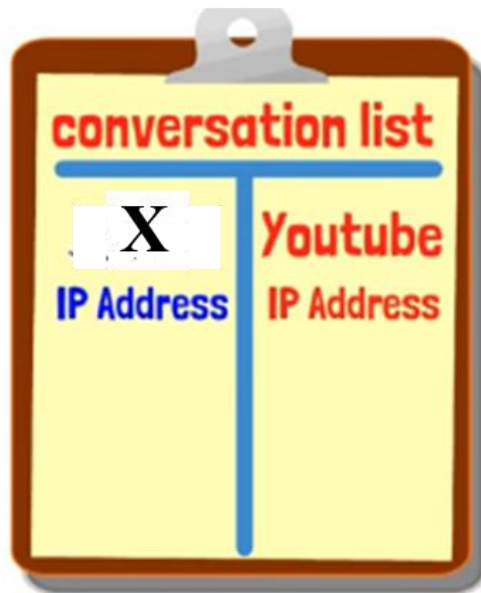
Access control  
list



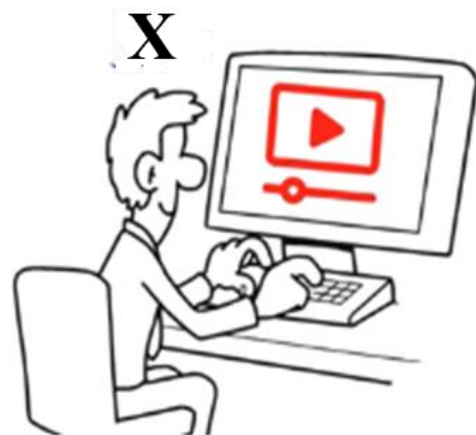


**Stateful Inspection Firewall**

**(More secure)**

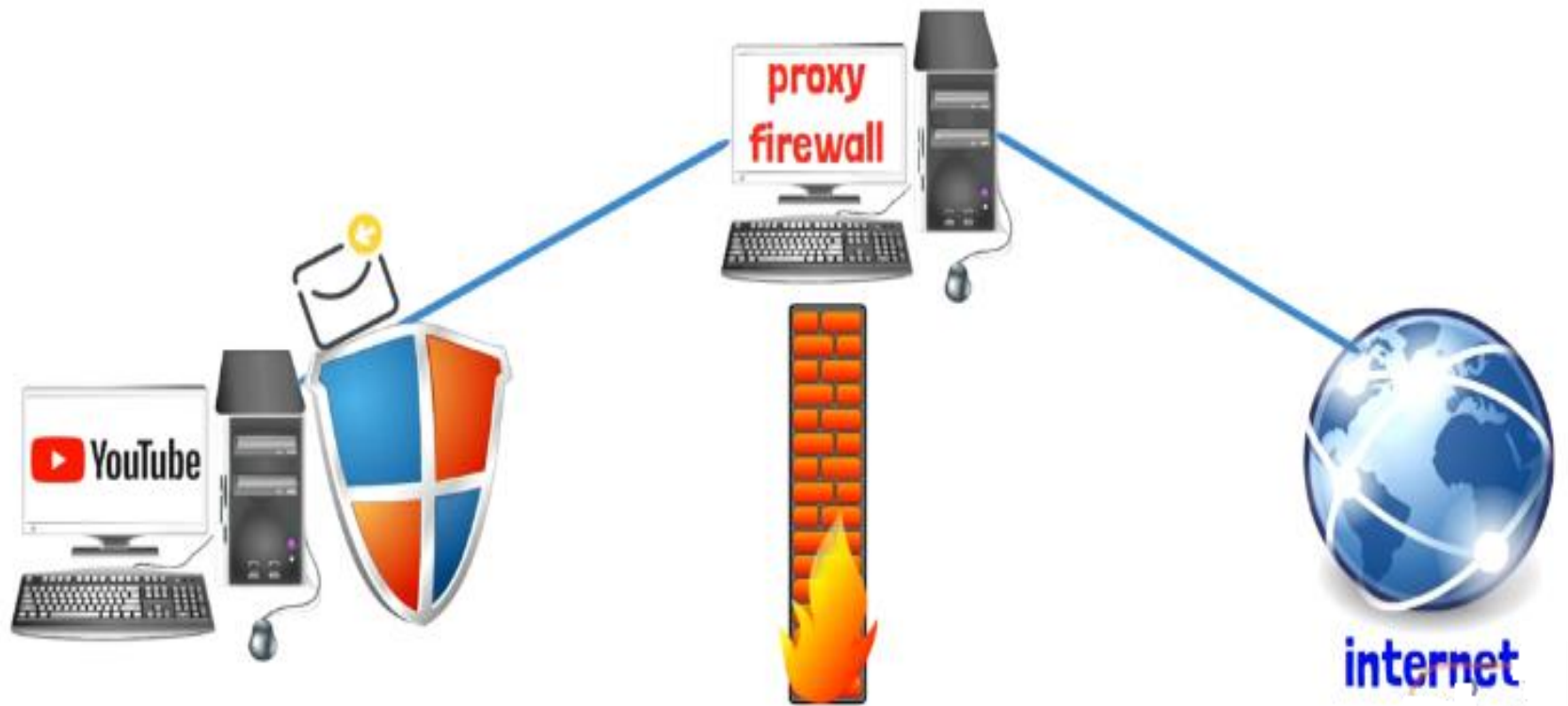


**Attack**



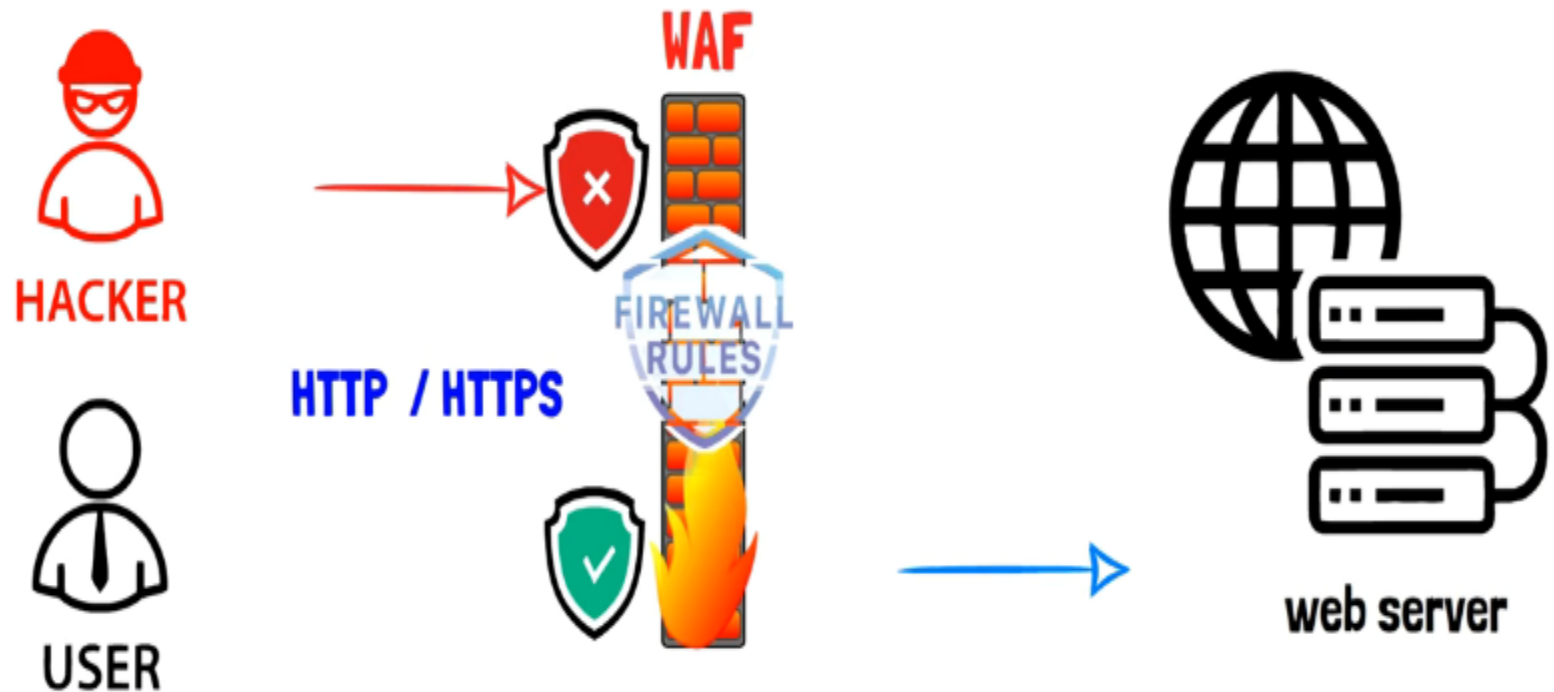
**YouTube**

# proxy firewall





## Web Application Firewall (WAF)



# Next-Generation Firewall

NGFW

## URL Filtering & Web Filtering



## (DPI) Deep Packet Inspection



## Intrusion Prevention System (IPS)



# Intrusion Prevention System (IPS)

(IPS)

